

# Demos of tables with `kable()` in knitr

Yihui Xie

September 28, 2026

The function `kable()` is a simple table generator. Unlike sophisticated packages (e.g. `xtable`, which has more than 30 arguments), this function only has a minimal number of arguments to generate simple tables.

```
library(knitr)
kable(head(iris))
```

| Sepal.Length | Sepal.Width | Petal.Length | Petal.Width | Species |
|--------------|-------------|--------------|-------------|---------|
| 5.1          | 3.5         | 1.4          | 0.2         | setosa  |
| 4.9          | 3.0         | 1.4          | 0.2         | setosa  |
| 4.7          | 3.2         | 1.3          | 0.2         | setosa  |
| 4.6          | 3.1         | 1.5          | 0.2         | setosa  |
| 5.0          | 3.6         | 1.4          | 0.2         | setosa  |
| 5.4          | 3.9         | 1.7          | 0.4         | setosa  |

```
kable(head(iris, 0))
```

| Sepal.Length | Sepal.Width | Petal.Length | Petal.Width | Species |
|--------------|-------------|--------------|-------------|---------|
|--------------|-------------|--------------|-------------|---------|

```
kable(head(iris, 1))
```

| Sepal.Length | Sepal.Width | Petal.Length | Petal.Width | Species |
|--------------|-------------|--------------|-------------|---------|
| 5.1          | 3.5         | 1.4          | 0.2         | setosa  |

## 1 No row names

```
kable(head(iris), row.names = FALSE)
```

| Sepal.Length | Sepal.Width | Petal.Length | Petal.Width | Species |
|--------------|-------------|--------------|-------------|---------|
| 5.1          | 3.5         | 1.4          | 0.2         | setosa  |
| 4.9          | 3.0         | 1.4          | 0.2         | setosa  |
| 4.7          | 3.2         | 1.3          | 0.2         | setosa  |
| 4.6          | 3.1         | 1.5          | 0.2         | setosa  |
| 5.0          | 3.6         | 1.4          | 0.2         | setosa  |
| 5.4          | 3.9         | 1.7          | 0.4         | setosa  |

## 2 At most 3 digits

```
kable(matrix(rnorm(20), 4), digits = 3)
```

|        |        |        |        |        |
|--------|--------|--------|--------|--------|
| -0.560 | 0.129  | -0.687 | 0.401  | 0.498  |
| -0.230 | 1.715  | -0.446 | 0.111  | -1.967 |
| 1.559  | 0.461  | 1.224  | -0.556 | 0.701  |
| 0.071  | -1.265 | 0.360  | 1.787  | -0.473 |

## 3 Alignment

By default, numeric columns are right-aligned, and other columns are left-aligned.

```
kable(head(iris), align = c("l", "c", "r", "l", "r"))
```

| Sepal.Length | Sepal.Width | Petal.Length | Petal.Width | Species |
|--------------|-------------|--------------|-------------|---------|
| 5.1          | 3.5         | 1.4          | 0.2         | setosa  |
| 4.9          | 3.0         | 1.4          | 0.2         | setosa  |
| 4.7          | 3.2         | 1.3          | 0.2         | setosa  |
| 4.6          | 3.1         | 1.5          | 0.2         | setosa  |
| 5.0          | 3.6         | 1.4          | 0.2         | setosa  |
| 5.4          | 3.9         | 1.7          | 0.4         | setosa  |

## 4 Default tables are ugly

Let's use the **booktabs** package. The air is suddenly fresh!

```
kable(head(iris), row.names = FALSE, booktabs = TRUE)
```

| Sepal.Length | Sepal.Width | Petal.Length | Petal.Width | Species |
|--------------|-------------|--------------|-------------|---------|
| 5.1          | 3.5         | 1.4          | 0.2         | setosa  |
| 4.9          | 3.0         | 1.4          | 0.2         | setosa  |
| 4.7          | 3.2         | 1.3          | 0.2         | setosa  |
| 4.6          | 3.1         | 1.5          | 0.2         | setosa  |
| 5.0          | 3.6         | 1.4          | 0.2         | setosa  |
| 5.4          | 3.9         | 1.7          | 0.4         | setosa  |

```
kable(head(iris, 0), row.names = FALSE, booktabs = TRUE)
```

| Sepal.Length | Sepal.Width | Petal.Length | Petal.Width | Species |
|--------------|-------------|--------------|-------------|---------|
|--------------|-------------|--------------|-------------|---------|

```
kable(head(iris, 1), row.names = FALSE, booktabs = TRUE)
```

| Sepal.Length | Sepal.Width | Petal.Length | Petal.Width | Species |
|--------------|-------------|--------------|-------------|---------|
| 5.1          | 3.5         | 1.4          | 0.2         | setosa  |

```
kable(head(mtcars[, 1:7]), booktabs = TRUE)
```

|                   | mpg  | cyl | disp | hp  | drat | wt    | qsec  |
|-------------------|------|-----|------|-----|------|-------|-------|
| Mazda RX4         | 21.0 | 6   | 160  | 110 | 3.90 | 2.620 | 16.46 |
| Mazda RX4 Wag     | 21.0 | 6   | 160  | 110 | 3.90 | 2.875 | 17.02 |
| Datsun 710        | 22.8 | 4   | 108  | 93  | 3.85 | 2.320 | 18.61 |
| Hornet 4 Drive    | 21.4 | 6   | 258  | 110 | 3.08 | 3.215 | 19.44 |
| Hornet Sportabout | 18.7 | 8   | 360  | 175 | 3.15 | 3.440 | 17.02 |
| Valiant           | 18.1 | 6   | 225  | 105 | 2.76 | 3.460 | 20.22 |

## 5 Long tables

```
kable(mtcars[, 1:7], longtable = TRUE, booktabs = TRUE)
```

|                     | mpg  | cyl | disp  | hp  | drat | wt    | qsec  |
|---------------------|------|-----|-------|-----|------|-------|-------|
| Mazda RX4           | 21.0 | 6   | 160.0 | 110 | 3.90 | 2.620 | 16.46 |
| Mazda RX4 Wag       | 21.0 | 6   | 160.0 | 110 | 3.90 | 2.875 | 17.02 |
| Datsun 710          | 22.8 | 4   | 108.0 | 93  | 3.85 | 2.320 | 18.61 |
| Hornet 4 Drive      | 21.4 | 6   | 258.0 | 110 | 3.08 | 3.215 | 19.44 |
| Hornet Sportabout   | 18.7 | 8   | 360.0 | 175 | 3.15 | 3.440 | 17.02 |
| Valiant             | 18.1 | 6   | 225.0 | 105 | 2.76 | 3.460 | 20.22 |
| Duster 360          | 14.3 | 8   | 360.0 | 245 | 3.21 | 3.570 | 15.84 |
| Merc 240D           | 24.4 | 4   | 146.7 | 62  | 3.69 | 3.190 | 20.00 |
| Merc 230            | 22.8 | 4   | 140.8 | 95  | 3.92 | 3.150 | 22.90 |
| Merc 280            | 19.2 | 6   | 167.6 | 123 | 3.92 | 3.440 | 18.30 |
| Merc 280C           | 17.8 | 6   | 167.6 | 123 | 3.92 | 3.440 | 18.90 |
| Merc 450SE          | 16.4 | 8   | 275.8 | 180 | 3.07 | 4.070 | 17.40 |
| Merc 450SL          | 17.3 | 8   | 275.8 | 180 | 3.07 | 3.730 | 17.60 |
| Merc 450SLC         | 15.2 | 8   | 275.8 | 180 | 3.07 | 3.780 | 18.00 |
| Cadillac Fleetwood  | 10.4 | 8   | 472.0 | 205 | 2.93 | 5.250 | 17.98 |
| Lincoln Continental | 10.4 | 8   | 460.0 | 215 | 3.00 | 5.424 | 17.82 |
| Chrysler Imperial   | 14.7 | 8   | 440.0 | 230 | 3.23 | 5.345 | 17.42 |
| Fiat 128            | 32.4 | 4   | 78.7  | 66  | 4.08 | 2.200 | 19.47 |
| Honda Civic         | 30.4 | 4   | 75.7  | 52  | 4.93 | 1.615 | 18.52 |
| Toyota Corolla      | 33.9 | 4   | 71.1  | 65  | 4.22 | 1.835 | 19.90 |

|                  |      |   |       |     |      |       |       |
|------------------|------|---|-------|-----|------|-------|-------|
| Toyota Corona    | 21.5 | 4 | 120.1 | 97  | 3.70 | 2.465 | 20.01 |
| Dodge Challenger | 15.5 | 8 | 318.0 | 150 | 2.76 | 3.520 | 16.87 |
| AMC Javelin      | 15.2 | 8 | 304.0 | 150 | 3.15 | 3.435 | 17.30 |
| Camaro Z28       | 13.3 | 8 | 350.0 | 245 | 3.73 | 3.840 | 15.41 |
| Pontiac Firebird | 19.2 | 8 | 400.0 | 175 | 3.08 | 3.845 | 17.05 |
| Fiat X1-9        | 27.3 | 4 | 79.0  | 66  | 4.08 | 1.935 | 18.90 |
| Porsche 914-2    | 26.0 | 4 | 120.3 | 91  | 4.43 | 2.140 | 16.70 |
| Lotus Europa     | 30.4 | 4 | 95.1  | 113 | 3.77 | 1.513 | 16.90 |
| Ford Pantera L   | 15.8 | 8 | 351.0 | 264 | 4.22 | 3.170 | 14.50 |
| Ferrari Dino     | 19.7 | 6 | 145.0 | 175 | 3.62 | 2.770 | 15.50 |
| Maserati Bora    | 15.0 | 8 | 301.0 | 335 | 3.54 | 3.570 | 14.60 |
| Volvo 142E       | 21.4 | 4 | 121.0 | 109 | 4.11 | 2.780 | 18.60 |

---

## 6 Want more features?

No, that is all I have. You should turn to other packages for help. I'm not going to reinvent their wheels.